



Digital visibility, physical obscurity: Uncovering the location strategies of ghost kitchens in platform urbanism

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ARTICLE INFO

Keywords:

Digital-physical nexus
Urban foodscape
Platform urbanism
Planning support systems
Machine learning

ABSTRACT

Digitalization and platform economies are transforming the restaurant sector, giving rise to 'ghost kitchens'—delivery-only operations that remain physically obscure yet digitally prominent. Their hidden nature creates new forms of spatial ambiguity and raises challenges for regulatory oversight. This study maps the spatial distribution of ghost kitchens in urban environments, identifies their clustering patterns, analyses their location strategies, and compares them with traditional and hybrid restaurant models. Our findings reveal distinct spatial patterns of ghost restaurants in areas with relatively 'weak' commercial locations, challenging the conventional wisdom that physical visibility is essential for restaurant success. By examining locational drivers at both horizontal (neighborhood function and street network proximity) and vertical (floor level and building characteristics) scales, we reveal a delivery-oriented siting logic that trades storefront exposure for access and efficiency, extending restaurant location theory to platform-mediated services. For practice, the findings guide retailers, platforms, and public agencies by identifying where ghost kitchens are most likely to emerge and which built-environment attributes shape their siting, thereby supporting evidence-based policies and governance that balance innovation with consumer protection, food safety, and public health.

1. Introduction

Digitalization and platform economies are reshaping the restaurant sector, giving rise to 'ghost kitchens'—delivery-only operations that are digitally prominent yet physically obscure (Almunawar and Anshari, 2022; Kifle Abebe et al., 2023; Souza et al., 2022; Tsai et al., 2023). By decoupling market exposure from street visibility (Shapiro, 2023), platform intermediation allows these establishments to thrive in inconspicuous or peripheral sites that would be unviable for traditional restaurants (Zhong et al., 2024). This shift creates new forms of spatial ambiguity, raising challenges for regulatory frameworks governing food safety and consumer protection. Ghost kitchens sit at one end of a virtual–physical spectrum in the restaurant sector, with traditional brick-and-mortar establishments at the other and hybrid online-to-offline formats in between.

Existing research on restaurant geography has consistently shown that traditional restaurants cluster in visible, high-traffic areas, closely

tied to the built environment, pedestrian flows, and urban vitality (Han et al., 2024; Hu et al., 2025). However, these established theoretical frameworks have paid insufficient attention to emerging virtual establishments (e.g., ghost kitchens and virtual restaurants), which likely exhibit fundamentally different spatial distributions. These establishments deliberately forgo the locational virtues sought by conventional restaurants, instead prioritizing what might be termed 'weak' locations—low-cost warehouses, peripherally situated properties, and inconspicuous middle floors of commercial buildings that can accommodate high delivery traffic volumes while avoiding the premium rents associated with traditional restaurant districts (Shapiro, 2023). These location strategies represent a significant departure from the clustering behaviors observed in conventional urban foodscape, which typically gravitate toward neighborhood commercial centers and tourist attractions with high pedestrian visibility (Han et al., 2024).

The strategic obscurity of ghost kitchens also carries important implications for consumer experience and regulatory oversight. Digitally,

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<https://doi.org/10.1016/j.jretconser.2026.104766>

Received 5 May 2025; Received in revised form 25 August 2025; Accepted 3 February 2026
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these establishments appear as polished, curated storefronts within delivery platforms, indistinguishable from traditional restaurants. Physically, however, they remain hidden from public view, undermining mechanisms of natural surveillance and complicating inspection (Hill, 2020; Y.-A. Kim and Hipp, 2021). Conventional food-safety frameworks (designed for visible, street-facing premises) struggle to adapt to these digitally mediated yet physically obscured sites (Graham, 2020; Laheri et al., 2025). Current regulatory approaches remain largely reactive (Dai and Wu, 2023), heavily dependent on consumer complaints that typically terminate at the platform level rather than addressing the actual food production sites (Laheri et al., 2025; Shapiro, 2023; Zhao et al., 2021). This digital-physical disconnect not only obscures production realities from consumers but also raises pressing questions for urban governance, platform accountability, and public health.

Despite the growing prominence of ghost kitchens, empirical evidence on their location patterns remains limited. Prior studies on restaurant geography have emphasized traditional models, leaving the digital-only sector underexplored (Xu et al., 2023). Preliminary qualitative work has highlighted regulatory challenges and consumer concerns (Nield et al., 2024), but systematic, quantitative analyses of ghost kitchen distribution are rare. In particular, little is known about whether these establishments are randomly scattered or exhibit identifiable clustering, and how their locational drivers differ from those of traditional and hybrid restaurants. Addressing these questions requires new methods that integrate multi-source digital data with spatial and machine learning approaches capable of capturing both horizontal (neighborhood function, street-network proximity) and vertical (floor level, building characteristics) dimensions of location.

This study responds to these gaps by systematically examining the hidden geography of ghost kitchens in Shenzhen, China, one of the world's most digitally advanced food-delivery markets. Drawing on comprehensive platform data and a novel classification framework, we map the distribution of traditional, hybrid, and ghost kitchens, identify clustering patterns, and model their location strategies across multiple scales. Our analysis reveals how delivery-oriented operations invert conventional siting logics: trading visibility and agglomeration for logistical efficiency and targeted access to digitally mediated demand.

The contributions of this research are twofold. First, we extend restaurant location theory by demonstrating how platform intermediation enables a new spatial logic of 'digital visibility with physical obscurity,' which challenges long-standing assumptions about the primacy of storefront exposure. This provides a theoretical bridge between classical location theory and contemporary debates in platform urbanism. Second, we generate empirical insights with practical value for urban planning and consumer-service regulation. By uncovering where ghost kitchens are most likely to emerge and which built-environment factors shape their siting, our findings offer a foundation for evidence-based policies that balance innovation with food safety, consumer protection, and public health.

The remainder of this paper is structured as follows: Section 2 reviews literature on restaurant geography and digital food platforms; Section 3 details our methodological approach; Section 4 presents empirical findings on spatial distributions and location determinants; Section 5 discusses implications for urban theory and governance; and Section 6 concludes with key insights and future research directions.

2. Literature review

2.1. The taxonomy of restaurant operations in the digital era

The restaurant industry has historically been characterized by its physical presence and direct customer interaction. However, the digital transformation of the food service sector has fundamentally altered traditional business models (Keeble et al., 2023). The rise of online food delivery platforms, such as Uber Eats, DoorDash, Grubhub, and

Meituan (Keeta) has created new market opportunities and challenges for restaurants (Azzoni et al., 2025). These platforms have enabled restaurants to reach customers beyond their immediate geographic vicinity, effectively expanding their market reach while simultaneously introducing new competitive dynamics (Dai and Wu, 2023; Feldman et al., 2023).

The terminology surrounding delivery-focused food operations, 'ghost kitchen,' 'cloud kitchen,' 'dark kitchen,' and 'virtual restaurant' lacks standardization across industry and academic literature, creating conceptual ambiguity despite their growing prominence in urban foodscapes. Shapiro (2023) defines ghost kitchens as food preparation facilities without storefronts or dine-in options, designed exclusively for delivery operations, a definition that emphasizes their complete detachment from traditional customer-facing functions. Similarly, Kulshreshtha and Sharma (2022) characterize cloud kitchens as 'commercial cooking spaces and take-aways without dining ... where geographical location is not much taken into consideration,' though they note the presence of 'parking and waiting areas,' suggesting minimal physical customer interaction. The term 'dark kitchen,' employed by Vu et al. (2023) and Li et al. (2020), follows this pattern, describing 'restaurants that do not have physical space for their patrons, mainly selling food through delivery services.' These three terms, ghost kitchen, cloud kitchen, and dark kitchen, can be considered functionally equivalent in their positioning on the physical-virtual spectrum, as they all operate without direct physical consumer exposure, mediating customer relationships exclusively through digital platforms (Laheri et al., 2025).

Virtual restaurants represent a more hybrid model within this taxonomy. Unlike purely virtual operations, they may exist as digital-only brands operating within established physical restaurants, leveraging existing kitchen infrastructure to produce multiple culinary concepts simultaneously (Zhao et al., 2021). This arrangement places them in an intermediate position on the virtuality spectrum—maintaining some physical presence while significantly depending on digital channels, with customers retaining the option for in-person visits despite the primary focus on delivery. At the opposite end of this continuum are traditional brick-and-mortar establishments that operate without online delivery services, maintaining complete physical presence and no virtual operations (Talamini et al., 2022). Rather than imposing rigid categorical distinctions that may not reflect operational realities, this research adopts a spectrum-based classification framework that acknowledges the varying degrees of virtuality and physicality in contemporary restaurant operations.

2.2. Location theory and spatial distribution of the restaurant industry

Restaurant location theory has evolved from classical economic geography principles like central place theory, which examined clustering patterns, to a more comprehensive conceptualization of strategic positioning factors (Gao et al., 2025). As digitalization progresses, significant shifts in the spatial preferences of commercial establishments have been observed. Built environment characteristics (e.g., building morphology, network configuration), socio-demographic factors, urban functions and competition are commonly investigated indicators used to identify differences between emerging and traditional business models (Gao et al., 2025; Yang et al., 2019). However, existing research predominantly focuses on individual business types in isolation, while comparative analyses between different operational business models, particularly between online and offline channels, remain limited (Xu et al., 2023). Consequently, the empirical impact of digitalization on the commercial landscape remains insufficiently documented, with many hypothesized effects lacking substantive evidence-based validation.

In the restaurant context, these foundational theories were adapted to explain the common clustering behavior observed in urban dining districts, where establishments benefit from agglomeration

economies—shared customer pools, reduced search costs for consumers (Y. Kim et al., 2024). Over time, this theoretical framework expanded to incorporate more nuanced understandings of how restaurants select locations based on a complex interplay of demographic, economic, and built environment variables.

Traditional location theory in the restaurant industry emphasizes the importance of foot traffic, visibility, accessibility, and proximity to target demographics (Hu et al., 2025). The classical mantra of ‘location, location, location’ has dominated strategic thinking in the sector for decades, with empirical studies consistently demonstrating strong correlations between restaurant performance and spatial attributes such as corner locations, street-level visibility, and proximity to complementary businesses or attractions (Han et al., 2024). The detailed location selection process often includes considerations from both the consumer perspective (convenience and visibility) and management perspective, such as rent costs and operational efficiency (Alwedyan, 2024; Meng et al., 2026). This dual-focused approach has guided restaurant entrepreneurs in identifying optimal locations that balance customer accessibility with business viability.

However, the emergence of ghost kitchens and virtual restaurants might fundamentally challenge these established principles. Their operations can theoretically function in areas with lower visibility and less commercial appeal (i.e., relatively weak location) while consumers can order food via online platforms without expending effort on physically searching for restaurant locations (Talamini et al., 2022). Vu et al. (2023) argue that for ghost kitchens, strategic positioning in high-traffic or high-visibility areas is less important than logistical efficiency that enables quick delivery and operational simplicity. Traditional good location thus becomes a potentially dispensable factor for ghost kitchens, contrasting sharply with its critical importance for conventional operations.

Despite this apparent freedom from traditional location constraints, recent studies suggest that spatial factors remain significant, albeit in different ways. Shapiro (2023) found that ghost kitchens tend to be located in areas with high population density, strong delivery infrastructure, and proximity to multiple delivery zones. It suggests a transformation rather than elimination of spatial dependencies. Some of these operations have been observed located in office buildings at an average floor of 13 for cheaper rent. While those platform-dependent restaurants were more dispersed both horizontally and vertically throughout urban areas (Talamini et al., 2022). This emerging spatial pattern represents not an absence of geographic logic but rather a reconfiguration of locational priorities around delivery efficiency rather than customer visibility. The location preferences of self-organized ghost kitchens are consistent with the collectively operated Cloud Kitchens run by Uber. They prioritize delivery radius coverage and operational efficiency over traditional retail considerations (Shapiro, 2023).

Although existing literature has summarized the location attributes that ghost kitchens may be inclined to select, there still remains a lack of empirical evidence to support these propositions. The ghost kitchens remain hidden within urban environments while their virtual presence significantly alters local industry geography and urban foodscape through digital platforms (Graham, 2020; He et al., 2019). This research gap requires moving beyond merely mapping ghost kitchens to make them ‘transparent’ (Fields et al., 2020). More importantly, we need frameworks that engage with what Bucher (2016) termed the ‘messiness’ of black-box algorithms in platform urbanism by examining hidden geographical logic shaping their spatial patterns.

2.3. Methodological approaches to studying virtual-physical restaurant spectrums

Existing studies on ghost kitchens and virtual restaurants have primarily utilized qualitative methodological approaches, revealing significant knowledge gaps that quantitative spatial analyses could address.

These studies have employed various methods, including surveys and interviews with restaurant operators (Vu et al., 2023), consumers (Nield et al., 2024), and the local authorities (Laheri et al., 2025). They have revealed significant challenges in regulatory oversight, particularly in inspecting and even locating ghost kitchens due to their inherent physical invisibility in the urban landscape. Consumer awareness remains strikingly low, with less than one-fourth of survey participants reporting familiarity with ghost kitchens, and fewer than 10% knowingly purchasing from them. Notably, most consumers express a desire for transparency regarding the source of their food purchases.

These preliminary qualitative explorations demonstrate a public desire for greater transparency in the urban foodscape, enabling more informed decision-making and addressing concerns about food safety standards (Nield et al., 2024). However, a substantial research gap exists in quantitative and spatial analysis that could leverage emerging data sources and analytical techniques to develop comprehensive classifications along the virtual-physical spectrum.

Recent advances in GeoAI fusing geospatial analysis with artificial intelligence (AI), such as large language models (LLMs), and computer vision, offer new opportunities for multi-dimensional analyses of restaurant operations (Keeble et al., 2023; Zhang et al., 2023). These technologies enable researchers to process diverse data types, including granular location information (down to building floor level) and visual representations of restaurant establishments. This technological capability provides a more holistic view of virtual-physical positioning within the foodscape and illuminates the spatial logic underlying these emerging restaurant models that often remain hidden from public view.

3. Methodology

3.1. Research design and conceptual framework

This study employs a quantitative analytical approach to differentiate location patterns and the underlying location factors among restaurants across the virtual-physical spectrum. The research is theoretically grounded in business location theory from economic geography, which provides established frameworks for understanding spatial decision-making. Our conceptual framework positions restaurants along a continuous spectrum rather than in binary categories, acknowledging that many establishments exhibit hybrid characteristics that blend traditional and virtual operational models. The methodology integrates three key technical components: multi-source data collection, comprehensive geographical analysis, and identification of distinct location strategies that reveal the spatial logic of restaurants at various points along the virtual-physical continuum. This integrated approach enables us to uncover previously hidden geographical patterns in the emerging urban foodscape (Fig. 1).

3.2. Restaurant data sources and collection

This study draws upon a comprehensive dataset of Point of Interest (POI) data extracted from three dominant platforms in China's restaurant ecosystem (desserts, snacks and drinks are excluded): Dianping (n = 141,859), Meituan (n = 63,582), and Ele.me (n = 45,076). These figures represent the final sample after rigorous deduplication procedures were applied to ensure data integrity. Collectively, these three platforms constitute the most extensive coverage of restaurant establishments across both physical and digital domains within the Chinese market.

Dianping, established in 2003, functions primarily as a local information and review platform where consumers share dining experiences and reporting and validating restaurants information. Each record includes a platform-verified ‘operational status’ field (*normal*, *closed down*, *duplicate*, *abnormal*, *not yet operating*). To ensure representativeness and avoid introducing noise from ephemeral or inactive establish-

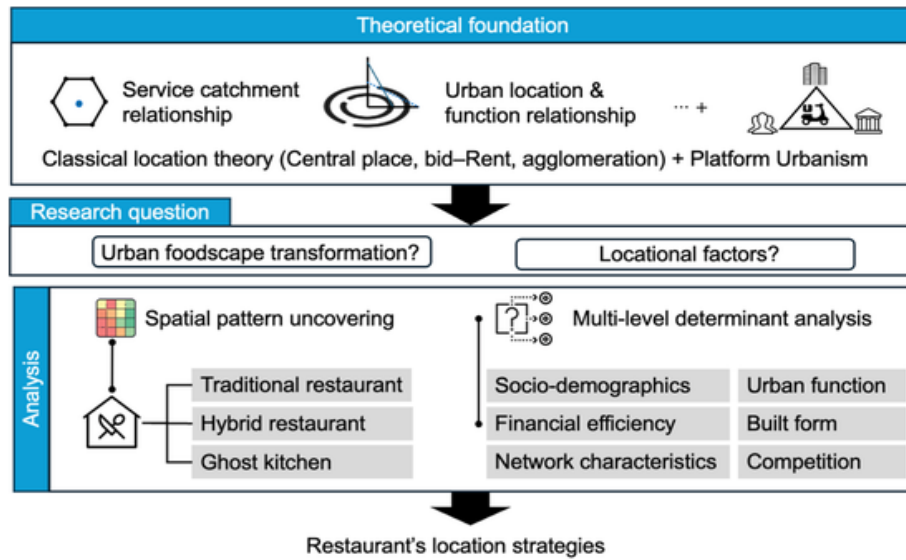


Fig. 1. Conceptual framework and research workflow.

ments, we excluded 8111 entries with any status other than *normal*, resulting in 166,360 active and verifiable establishments. Further filtering removed dessert, snack, and beverage-only outlets that typically lack dine-in facilities, yielding a final Dianping sample of 141,859 establishments. Meituan and Ele.me, founded in 2010 and 2008 respectively, operate as specialized food delivery platforms that collectively command the majority of China's online food delivery market share (Dai and Wu, 2023). This market duopoly ensures near-complete coverage of digitally-accessible food service establishments. Raw datasets from these two platforms underwent a rigorous deduplication process both within each platform and across platforms to remove repeated listings. Invalid or non-operational entries were also excluded to match the quality standards applied to Dianping data.

All these data were systematically collected during September 2024 using standardized web scraping protocols to maintain consistency across platforms. The temporal alignment of data collection across all three sources minimizes potential discrepancies due to market fluctuations or seasonal variations, thus enhancing the reliability of cross-platform data fusion and comparison.

3.3. Data source for measuring urban metrics

The POI data representing other urban functions is collected from Amap (<https://lbs.amap.com/api/webservice/>). As for the population-related information, we use the Baidu Huiyan Location-based services data (<https://lbsyun.baidu.com/>) (hereafter LBS) to calculate the socio-demographic factors that also represent the potential market demand. LBS monitors both the working and resident population at the 100-m grid level. OSM Road Network (<https://www.openstreetmap.org/>) to calculate network indicators (e.g., betweenness centrality, intersection and road network density). Building data was collected from Baidu Map (<https://map.baidu.com/>) from Dec 2024 with building height information emerging with the fields of function and year of completion in recently published dataset CMAB (Zhang et al., 2025). Rent of each land parcel was collected from Open Data Platform (<https://opendata.sz.gov.cn/>) operated by the Shenzhen Government for financial efficiency estimation.

3.4. Virtual-physical spectrum classification

This study implements the Restaurant Identity Modes Spectrum (RIMS) framework (Deng et al., 2026), a novel methodological approach for classifying food service establishments along the virtual-physical continuum. RIMS employs a sophisticated multi-source triangulation technique that integrates and analyzes heterogeneous data from various digital platforms to categorize restaurants into three distinct operational archetypes: traditional brick-and-mortar establishments (hereafter traditional restaurant), which maintain a primarily physical presence; online-to-offline operations (hereafter hybrid restaurant), which strategically leverage both physical and digital channels; and virtual-only establishments (hereafter ghost kitchen), also commonly known as ghost or cloud kitchens, which operate exclusively through digital interfaces without conventional dining facilities (Fig. 2). The classification procedure consists of three sequential identification processes.

3.4.1. Physical presence detection

The initial classification phase utilized the Dianping dataset as the authoritative source for identifying traditional brick-and-mortar establishments. Restaurants in this dataset were categorized as physical establishments due to Dianping's verification procedures that confirm the existence of physical storefronts (establishments without comment and rating are excluded). These establishments represent the traditional or hybrid of the RIMS continuum, characterized by the availability of in-store dining options.

3.4.2. Hybrid channel matching

The second phase identified online-to-offline (O2O) operations through a sophisticated proximity-based correlation algorithm applied to the Meituan and Eleme delivery platform datasets. For each restaurant in these delivery platforms, we employed spatial proximity analysis to identify the 20 nearest potential matches from unclassified Dianping establishments, creating candidate pairs for further analysis. We then implemented a multi-modal similarity assessment framework combining textual and visual verification methods. The textual component utilized a large language model (LLM) to calculate similarity scores between establishment names and addresses, while the visual component employed a vision language model (VLM) to analyze logo images for identity confirmation through visual similarity analysis. To

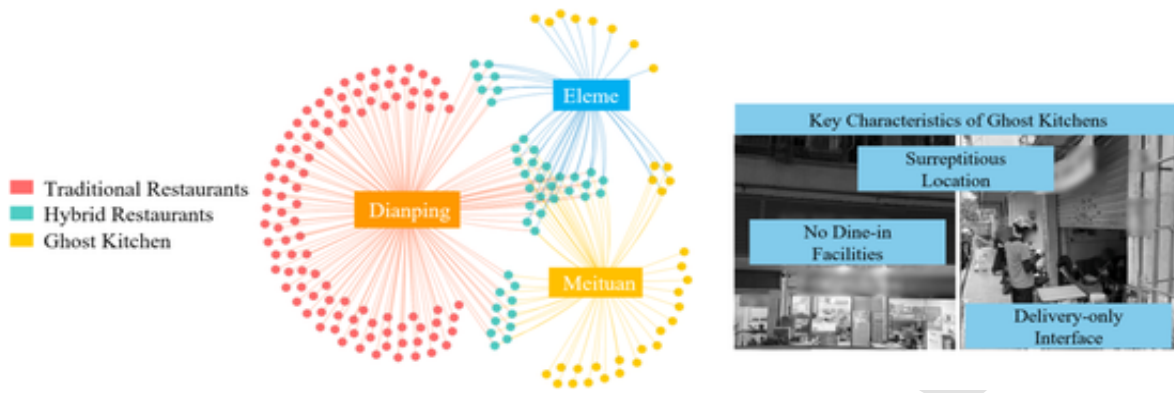


Fig. 2. The schematic Venn Network diagram of the classification in this study.

ensure robust classification, we developed a supervised learning model using XGBoost, trained on over 2500 manually labeled establishment pairs, achieving excellent performance metrics (F1 score: 0.96, Accuracy: 0.94). This model effectively distinguishes between genuine matches and pseudo-matches, allowing us to reliably identify hybrid establishments that maintain both physical storefronts and digital delivery presence.

3.4.3. Virtual-only identification

The final classification phase employed a systematic residual analysis approach to identify ghost kitchens. After accounting for traditional and hybrid establishments, we applied our multi-step matching technique and pre-trained model to the remaining unmatched entities in the Meituan and Eleme datasets to eliminate potential duplicates. From this refined pool of candidates, we selected a stratified random sample for comprehensive manual verification, examining each establishment for: (1) confirmed absence of corresponding physical locations in the Dianping dataset; (2) operational characteristics distinctive of virtual-only establishments; and (3) specific identifiers indicating purpose-built delivery-only operations. This rigorous, multi-stage classification methodology enabled us to comprehensively map the full spectrum of restaurant virtualization within the urban environment, establishing a robust taxonomic foundation for our subsequent spatial analysis of ghost kitchen distribution patterns.

3.5. Spatial analysis methods

3.5.1. Spatial distribution patterns

To systematically examine the hidden geography of ghost kitchens, we employed hotspot analysis using the Getis-Ord G_i^* statistic across 500-m grid cells. This method identifies statistically significant spatial clusters of high values (hotspots) and low values (coldspots) for each restaurant type. We then conducted comparative analysis between traditional restaurants, hybrid models, and ghost kitchens to identify distinctive locational preferences.

3.5.2. Multi-level determinant analysis

Our research employed a multi-scale analytical framework to comprehensively investigate the locational determinants of ghost kitchens across the urban landscape. At the macro scale, we applied Ordinary Least Squares (OLS) regression to 500-m grid cells, using the count of each restaurant type (traditional, hybrid, ghost kitchen) as the dependent variables. Independent variables comprised grid-level metrics representing the built environment, urban functional characteristics, street network properties, and socio-demographic conditions. At the micro scale, we implemented an establishment-level XGBoost binary classification model, where the dependent variable was whether a restaurant

was a ghost kitchen. Independent variables included finer-resolution (100-m buffer) urban metrics, building-specific attributes, and establishment-level characteristics. This dual-scale approach allowed us to identify both broad spatial trends and detailed vertical/horizontal positioning strategies relevant to ghost kitchens.

3.5.3. Factor selection

Independent variables were assembled to reflect five hypothesized channels: (i) built form and development intensity (Wu et al., 2021), (ii) urban functions (Yang et al., 2019), (iii) network structure and delivery access, (iv) financial efficiency (rent) (Wu et al., 2021), (v) socio-demographics structure (residents, employees, age, education, income at 100-m grid and 2 KM catchments) (Hu et al., 2025), and (vi) competitive environment (HHI) (Gao et al., 2025). Table 1 lists variables, summary statistics, and the model(s) where each enters. The detail model specification and factor selection justification are included in Appendix 1.

4. Results and discussion

4.1. Spatial pattern of restaurants along the virtual-physical spectrum

Based on the Restaurant Identity Modes Spectrum (RIMS) developed for this study, we analyzed the spatial distribution of traditional restaurants, hybrid restaurants, and ghost kitchens across Shenzhen. Our September 2024 dataset identified 156,163 unique restaurant points of interest (POIs), distributed along the virtual-physical continuum as follows: 91,000 traditional restaurants (58%) with established physical presence, 51,861 hybrid establishments (33%) operating through both physical and virtual channels, and 13,302 ghost kitchens (9%) functioning primarily visible in the digital realm.

The 500-m grid aggregation of these POIs reveals distinct hot and cold spots for each restaurant type (Fig. 3). Traditional and hybrid restaurants exhibit remarkably similar spatial patterns, with well-defined hot spot clusters in urban core areas. Ghost kitchens, despite their smaller numbers, follow a comparable spatial distribution, with particularly strong concentrations in central business districts such as Shekou in the southwestern part of the city. Overall, the hot spot analysis indicates that all three restaurant types tend to cluster in Shenzhen's central-southern urban areas (historically known as 'Guangnei', or 'inside the pass', dominated by white-collar employment) with significant hot spots in these regions. In contrast, cold spots predominantly appear in the northern districts (the 'outside the pass' areas), which are characterized by industrial landscapes and manufacturing activities (Cho, 2017; Lin and While, 2022).

Fig. 4 presents a comparative hot spot analysis that highlights the nuanced differences in spatial distribution among the three restaurant

Table 1

Variable summary and model application.

Variable	Abbreviation	Mean	Std Dev	Macro	Micro
Dependent variables					
No. of traditional restaurants	TradRest	18.661	33.334	✓	×
No. of hybrid restaurants	HybridRest	15.907	22.388	✓	×
No. of ghost kitchens	GKit	4.553	7.946	✓	×
Is ghost kitchen (binary classification)	isGkit	-	-	×	✓
Independent variables					
Building characteristics					
Building count	BldgCnt	190.711 (18.596)	154.619 (14.517)	✓	✓
Average building height (m)	BldgHgt	18.201 (23.958)	8.767 (14.803)	✓	✓
Building compactness	BldgComp	0.604 (0.603)	0.106 (0.08)	✓	✓
Building coverage ratio	BCR	0.217 (0.322)	0.099 (0.142)	✓	✓
Floor area ratio	FAR	1.504 (2.688)	0.860 (2.062)	✓	✓
Pervious surface coverage	PSC	0.375 (0.048)	0.105 (0.018)	✓	✓
Year of Completion of building located	YOC	(1995)	(7.386)	×	✓
Floor of restaurant in building	Floor	(0.204)	(1.339)	×	✓
Urban functions					
POI count (No./Hectare)	POICnt	171.992 (10.481)	191.551 (16.709)	✓	✓
POI diversity	POIDiv	1.678 (0.776)	0.562 (0.714)	✓	✓
Urban location categories (1 = urban center, 2 = near suburbs & new towns, and 3 = far suburbs)	UrLocCat	-	-	✓	×
Urban village	UrVill	-	-	×	✓
Building function	BldgFtn	-	-	×	✓
Financial efficiency					
Average rent (RMB/Mon-Sqm)	Rent	43.399 (51.262)	33.318 (32.45)	✓	✓
Socio-demographic					
Residential population (1000 persons)	ResPop	6.289 (0.727)	5.128 (0.579)	✓	✓
No. of employees (1000 persons)	Emp	2.694 (0.262)	2.853 (0.382)	✓	✓
Proportion of residents aged below 55	Age < 55	0.979 (0.983)	0.021 (0.025)	✓	✓
Proportion of residents with a bachelor's degree or higher	Edu ≥ Bach	0.080 (0.077)	0.074 (0.072)	✓	✓
Proportion of residents with income higher than the city average	Inc ≥ Avg	0.317 (0.349)	0.125 (0.094)	✓	✓
Residential population in 2 KM catchment (1000 persons)	ResPop (2 KM)	(302.437)	(126.954)	×	✓
No. of employees (1000 persons)	Emp (2 KM)	(137.299)	(78.376)	×	✓
Proportion of residents aged below 55 in 2 KM catchment	Age < 55 (2 KM)	(0.976)	(0.02)	×	✓
Proportion of residents with a bachelor's degree or higher in 2 KM catchment	Edu ≥ Bach (2 KM)	(0.094)	(0.054)	×	✓
Proportion of residents with income higher than the city average in 2 KM catchment	Inc ≥ Avg (2 KM)	(0.349)	(0.094)	×	✓
Network characteristics					
Betweenness centrality	BtwCent	0.004 (0.002)	0.007 (0.005)	✓	✓

Table 1 (continued)

Distance of building located to the primary network (m)	DistPri	(34.311)	(33.161)	×	✓
Competition					
Herfindahl index	HHI	0.705	0.241	✓	×

Note.

1) Values in parentheses represent statistics calculated specifically for the micro-level model subsample.

2) '-' indicates not applicable for non-continuous variables.

3) ✓ indicates variable is included in the respective model; × indicates not included.

types. When comparing hybrid restaurants and ghost kitchens to traditional restaurants (Fig. 4a), we observe that the central (Minzhibantian), central-southern (Futian and Luohu) and southwestern (Qianhai) parts of Shenzhen maintain 'Hot to Hot' status (pink areas) across all restaurant types, confirming these as prime locations for food service regardless of business model. The northern regions (Pinghu, Longhua, and Pinshan) consistently show 'Cold to Cold' patterns (light blue), indicating lower restaurant density across all types. Notably, there are significant 'Cold to Not Sig' areas (blue cross-hatched) in the north and northeast (Ban'an, Yanluo, Guangming, and Longang), particularly pronounced in the ghost kitchen comparison (Fig. 4b). The 'Hot to Not Sig' areas (red crosshatched) appear prominently along the periphery of hot zones (Nanshan), suggesting transitional areas where hybrid restaurants and ghost kitchens have already emerged and clustered with higher density than traditional restaurants. The limited 'Not Sig to Hot' areas (solid red) indicate locations where traditional models still maintain particular traction.

4.2. Restaurants' vertical distribution

Beyond horizontal distribution, our analysis of vertical distribution patterns (Fig. 4) reveals significant differences among restaurant types. Traditional restaurants demonstrate the highest ground floor presence (89%), aligning with their need for street-level visibility and walk-in customers. In contrast, ghost kitchens exhibit greater flexibility in floor location, with only 79% on ground floors, 11% on first floors, and 5% in lower-ground/basement spaces—reflecting their delivery-focused model that requires less customer visibility. Hybrid restaurants position themselves between these patterns with 84% ground floor occupancy.

The vertical range also varies considerably, with traditional restaurants showing the greatest vertical extent (maximum: 98 floors) compared to hybrid restaurants (maximum: 39) and ghost kitchens (maximum: 13). This suggests traditional restaurants are more integrated into high-rise commercial and mixed-use developments, often leveraging scenic views (skyview or cloudview) as attractions—a selling point irrelevant to ghost kitchens.

These patterns reveal how restaurant models adopt different location strategies: traditional restaurants prioritize prime visibility locations and establish presence across the vertical dimension of the city, while ghost kitchens demonstrate greater location flexibility, concentrating in similar hot zones but willingly occupying non-ground floor spaces. Our analysis highlights how emerging restaurant models like ghost kitchens adapt their location strategies based on their operational requirements. By occupying spaces that would be disadvantageous for traditional restaurants (non-ground floors, less visible locations), ghost kitchens reduce rental costs while maintaining proximity to delivery markets.

4.3. Restaurants' location preference and explanation

4.3.1. Macro-level analysis: restaurant location across the urban landscape

Conventional location theory suggests that restaurants tend to establish in areas with favorable visual access, high foot traffic, and prox-

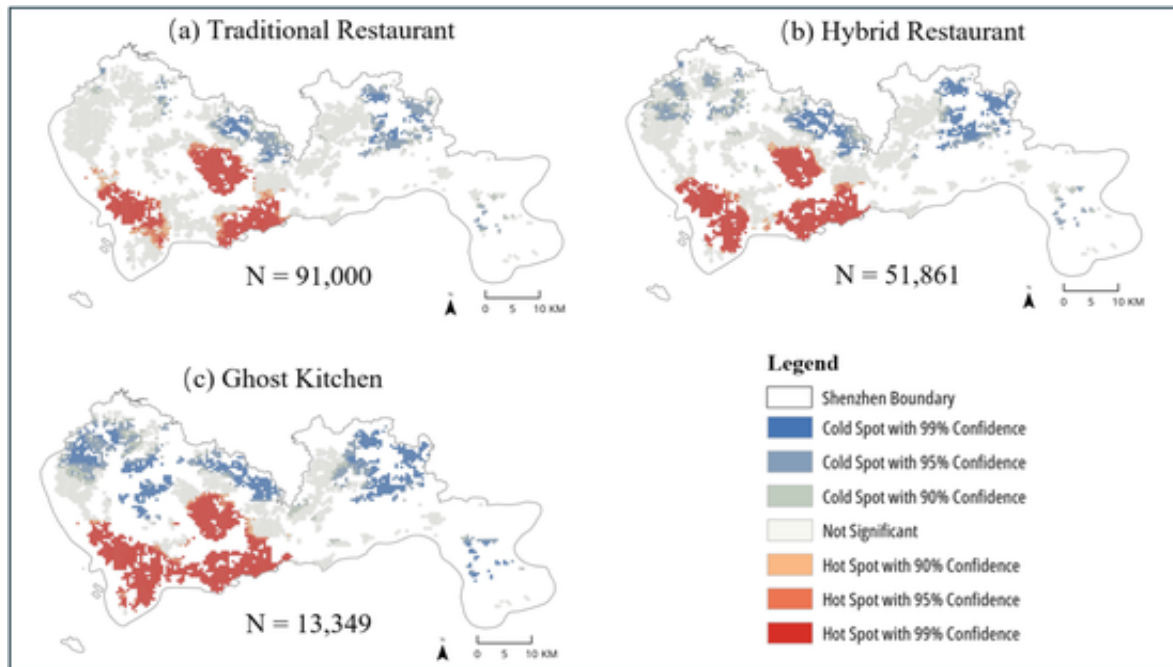


Fig. 3. Spatial distribution of hot spot and cold spot.

imity to markets. To examine how various urban metrics influence restaurant distribution, we analyzed multiple categories of variables across 3257,500-m grids in Shenzhen, each containing at least one restaurant. Our linear regression models reveal that these variables have differential impacts across three restaurant types: traditional restaurants, hybrid restaurants, and ghost kitchens (as shown in Table 2).

The built environment significantly influences restaurant distribution patterns. Building count demonstrates a consistent positive effect across all restaurant types, with traditional restaurants exhibiting the strongest relationship (coefficient = 0.011, $p < 0.01$), followed by hybrid restaurants (0.007, $p < 0.01$) and ghost kitchens (0.003, $p < 0.01$). This suggests that areas with more numerous buildings provide greater opportunities for restaurant establishment. Conversely, average building height negatively impacts all restaurant categories, with traditional restaurants showing the greatest sensitivity (coefficient = -0.290 , $p < 0.01$) compared to hybrid restaurants (-0.240 , $p < 0.01$) and ghost kitchens (-0.107 , $p < 0.01$). This finding indicates a preference for lower-rise environments for restaurant operation. Building coverage ratio demonstrates a significant negative effect only for ghost kitchens (-6.190 , $p < 0.01$), indicating these delivery-focused establishments may favor areas with lower building density, possibly due to operational considerations such as delivery logistics and cost efficiency. Notably, floor area ratio and pervious surface coverage showed no significant effects across all restaurant types, suggesting that these density and environmental measures are less critical in location decisions than previously hypothesized.

The intensity of urban functions, measured by POI count, consistently demonstrates a positive relationship across all models. Traditional restaurants exhibit the strongest association (coefficient = $8.6e-03$, $p < 0.01$), followed by hybrid restaurants ($7.9e-03$, $p < 0.01$) and ghost kitchens ($2.3e-03$, $p < 0.01$). This finding aligns with Han et al. (2024), who articulated that restaurants are more likely to locate near complementary attractions. However, POI diversity proved statistically insignificant across all models, suggesting that while the absolute concentration of points of interest matters, the variety of these attractions does not significantly influence restaurant location decisions.

Traditional restaurants demonstrate a preference for central, higher-rent locations, as evidenced by the significant negative effect of Urban location category (1 = urban center, 2 = near suburbs & new towns, and 3 = far suburbs) (coefficient = -1.442 , $p < 0.01$) and positive effect of average rent (0.044, $p < 0.001$). While the urban location effect is modest in magnitude, both findings align in suggesting that traditional restaurants are willing to pay premium prices for central locations, likely reflecting their dependence on foot traffic and visibility. In contrast, hybrid restaurants and ghost kitchens show no significant sensitivity to either spatial hierarchy or average rent levels, indicating more flexible location strategies that prioritize operational efficiency over premium real estate positioning.

Betweenness centrality, which measures how frequently a node serves as a bridge along shortest paths between other nodes in the network, did not demonstrate statistical significance in explaining the location patterns of any restaurant type. This suggests that positioning at key junctions or intersection points within the street network is not a primary determinant for restaurant location decisions, regardless of their business model. A more granular analysis of network characteristics at the individual establishment level, rather than the current 500-m grid resolution, may be necessary to accurately model location preferences.

Proximity to markets, represented by both residential population and the number of employees, correlates positively with restaurant numbers across all types. Residential population demonstrated the strongest effect on traditional restaurants (coefficient = 3.159, $p < 0.001$), compared to hybrid restaurants (2.389, $p < 0.001$) and ghost kitchens (0.685, $p < 0.001$). Similarly, the number of employees showed significant positive effects across all models: traditional (2.121, $p < 0.001$), hybrid (2.820, $p < 0.001$), and ghost kitchens (0.910, $p < 0.001$). Interestingly, hybrid restaurants show the highest sensitivity to the number of employees, possibly reflecting their dual role in serving both dine-in customers during business hours and delivery orders throughout the day.

The proportion of the population under 55 years (the young and middle-aged) demonstrates strong positive effects across all restaurant types, with traditional restaurants showing the highest coefficient

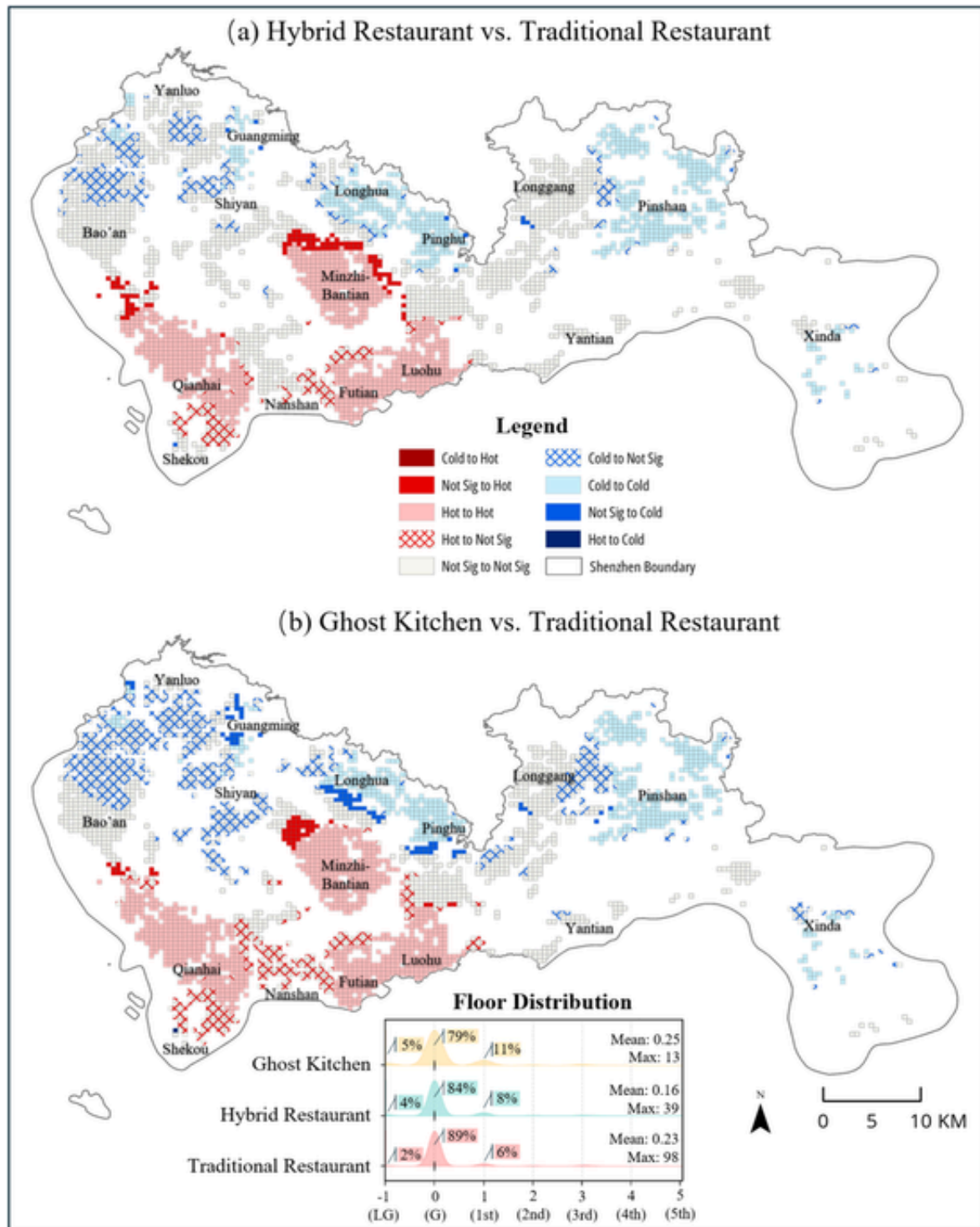


Fig. 4. Hotspot Analysis Comparison: (a) Hybrid Restaurant vs. Traditional Restaurant; (b) Ghost Kitchen vs. Traditional Restaurant.

(146.99, $p < 0.01$), followed by hybrid restaurants (46.648, $p < 0.01$) and ghost kitchens (25.767, $p < 0.01$). This reflects the younger demographic's higher propensity for dining out and using food delivery services. Educational attainment, measured by the proportion of bachelor's degree holders and higher, shows significant negative effects for traditional (-30.804 , $p < 0.01$) and hybrid restaurants (-31.822 , $p < 0.01$), while remaining non-significant for ghost kitchens. Conversely, the proportion of above-average income population shows pos-

itive effects across all models: traditional (13.927, $p < 0.01$), hybrid (26.983, $p < 0.01$), and ghost kitchens (6.231, $p < 0.01$), with hybrid restaurants demonstrating the strongest response to income levels.

The Herfindahl index, measuring market concentration and competition intensity, shows significant negative effects across all restaurant types. Traditional restaurants exhibit the strongest sensitivity to competition (-29.487 , $p < 0.01$), followed by ghost kitchens (-5.625 , $p < 0.01$) and hybrid restaurants (-3.959 , $p < 0.01$). This indicates

Table 2

OLS regression result examining the relationship between urban metrics and no. of restaurants by type.

Variable	Traditional restaurant	Hybrid restaurant	Ghost kitchen
Building characteristics			
BldgCnt	0.011*** (0.004)	0.007*** (0.003)	0.003** (0.001)
BldgHgt	-0.290*** (0.073)	-0.240*** (0.053)	-0.107*** (0.020)
BldgComp	-7.335* (4.094)	-2.844 (2.989)	0.494 (1.112)
BCR	-8.324 (8.524)	-7.886 (6.209)	-6.190*** (2.310)
FAR	-1.140 (1.168)	0.118 (0.852)	0.358 (0.317)
PSC	1.439 (3.742)	-0.469 (2.728)	0.029 (1.016)
Urban functions			
POICnt	0.009*** (0.002)	0.008*** (0.002)	0.002*** (0.001)
POIDiv	-1.397 (0.717)	0.524 (0.523)	-0.124 (0.195)
UrLocCat	-1.442** (0.688)	-0.331 (0.502)	-0.149 (0.187)
Financial efficiency			
Rent	0.044*** (0.017)	0.018 (0.012)	0.005 (0.005)
Socio-demographic			
ResPop	3.159*** (0.106)	2.389*** (0.074)	0.685*** (0.028)
Emp	2.1212*** (0.184)	2.820*** (0.133)	0.910*** (0.050)
Age < 55	146.99*** (28.732)	46.648*** (20.971)	25.767*** (7.798)
Edu ≥ Bach	-30.804*** (13.927)	-31.822*** (9.954)	-4.172 (3.705)
Inc ≥ Avg	13.927*** (7.943)	26.983*** (5.779)	6.231*** (2.152)
Network characteristics			
BtwCent	-26.419 (50.203)	-20.094 (36.626)	-9.999 (13.633)
Competition			
HHI	-29.487*** (1.101)	-3.959*** (1.100)	-5.625*** (0.364)
Model Statistics			
R-squared	0.63	0.57	0.52
Adjusted R-square	0.63	0.56	0.52
No. of 500-m Grid	3257	3257	3257

Note.

1) Values in parentheses are standard error.

2) Significance levels: ***p < 0.01, **p < 0.05, *p < 0.1.

that all restaurant types tend to avoid highly concentrated competitive environments, with traditional restaurants showing the greatest aversion to competition, possibly due to their reliance on local customer bases.

These findings align with restaurant location strategies observed in competitive online-to-offline food ordering scenarios studied by He et al. (2019). Notably, the principle of ‘the closer, the better’ remains effective even within the context of online platforms offering seemingly equalizing delivery services. The traditional restaurant model demonstrated the highest explanatory power (R-squared = 0.63), followed by the hybrid restaurant model (R-squared = 0.57), while the ghost kitchen model exhibited the lowest (R-squared = 0.52). This variance in explanatory power suggests that conventional location theories are most applicable to traditional restaurant models, moderately applicable to hybrid formats, while ghost kitchens operate under a different location logic that warrants further investigation with additional variables beyond traditional urban metrics.

4.3.2. Micro-level analysis: unmasking ghost kitchen location strategies

At the micro-level, we trained an establishment-level XGBoost classifier where ghost kitchens were the positive class and traditional plus hybrid restaurants formed a single comparison group. Given the scarcity of ghost kitchens relative to conventional restaurants, we applied the Synthetic Minority Over-sampling Technique (SMOTE) to achieve a balanced dataset. This technique generates synthetic samples for the minority class by interpolating between existing observations, a widely accepted approach in machine learning for imbalanced classification problems (Wang et al., 2021). The trained model demonstrated

strong classification performance with an overall accuracy of 0.89 and robust F1 scores for both categories.

Fig. 5 displays the feature importance scores and SHAP value derived from our XGBoost model, revealing that socio-demographical factors play crucial roles in differentiating ghost kitchens from other restaurants. Most notably, Inc ≥ Avg (2 KM), Edu ≥ Bach and Age < 55 (2 KM) emerge as the strongest predictors. This pattern aligns with our macro-level grid analysis. Following by the factors reflect the urban function, such as POIDiv, POICnt and UrVill. Rent reflecting the financial efficiency is also detected being an important feature for the perdition model. Micro-level factors representing the network position and accessibility, such as BtwCent (of the nearest node) and DistPri are also detected their importance on the prediction. Those features reflecting the physical landscape and could be distinguish from eye sight (BldgCnt, BldgHgt, FAR and BCR) were tested less important. Generally, those non-physical or non-directly observable features have comparatively higher importance on the perdition model.

Fig. 6 presents partial-dependence curves with 5–95% bootstrap confidence bands for key numeric predictors. They indicate three coherent location themes.

A) Social-demographic characteristics and targeted group

A higher share of residents with income above the city average in the 2 KM catchment lowers the probability of a ghost kitchen (Fig. 6a), while a higher share with a bachelor's degree or above raises it (Fig. 6c). Younger population shares, both at the grid scale and within the 2 KM catchment, show positive effects that taper at the extreme upper tail (Fig. 6d and g). Residential population displays a clear hump-shaped pattern, with higher probabilities at moderate neighborhood densities and attenuation in very sparse or very dense settings (Fig. 6l). Employment increases the probability at low values and then flattens, indicating diminishing returns relative to residential demand (Fig. 6n). Together these profiles suggest targeted positioning toward educated, mixed-age residential demand rather than uniformly high-spending markets, echoing recent location comparison study on traditional and new retail (e.g., Gao et al., 2025).

B) Functional characteristics and economic centrality

Ghost kitchen probability declines as POI diversity and POI count increase (Fig. 6b & f). This indicates that ghost kitchens employ location strategies that depart from traditional commercial spillover effects, which typically attract consumers through diversified functions that accommodate daily needs (Jiang et al., 2024). Furthermore, ghost kitchens demonstrate a higher likelihood of operating within residential buildings (class value 1 = positive; 0 = negative) (Fig. 5), aligning with their strategy to minimize rental costs while avoiding visibility in conventional commercial spaces. This evidence supports Laheri et al. (2025) findings that ghost kitchens prefer intentional invisibility, while the location impedes regular inspections, monitoring, and enforcement of food safety standards by local authorities.

Contrary to initial expectations, ghost kitchens demonstrate a higher probability of location in areas with elevated rent values (Fig. 6e). This pattern corresponds with the clustering identified in Fig. 2, showing hotspots in Shenzhen's economically developed southern region known as ‘inside the pass’ (Guangnei). Thus, land parcel rent functions more as a proxy for urban centrality rather than merely indicating financial efficiency. Also, the variables Urban village trends upward (Fig. 6i) and Year of completion trends downward (Fig. 6j), indicating urban village (usually cheaper) and older stock are more typical hosting fabric. At the city scale, these findings unveil their strategic approach to positioning in central-but-functionally-weaker micro-environments. This strategy evokes the Chinese proverb: ‘Real retreat is to hide among the crowd, not in the wilderness’.

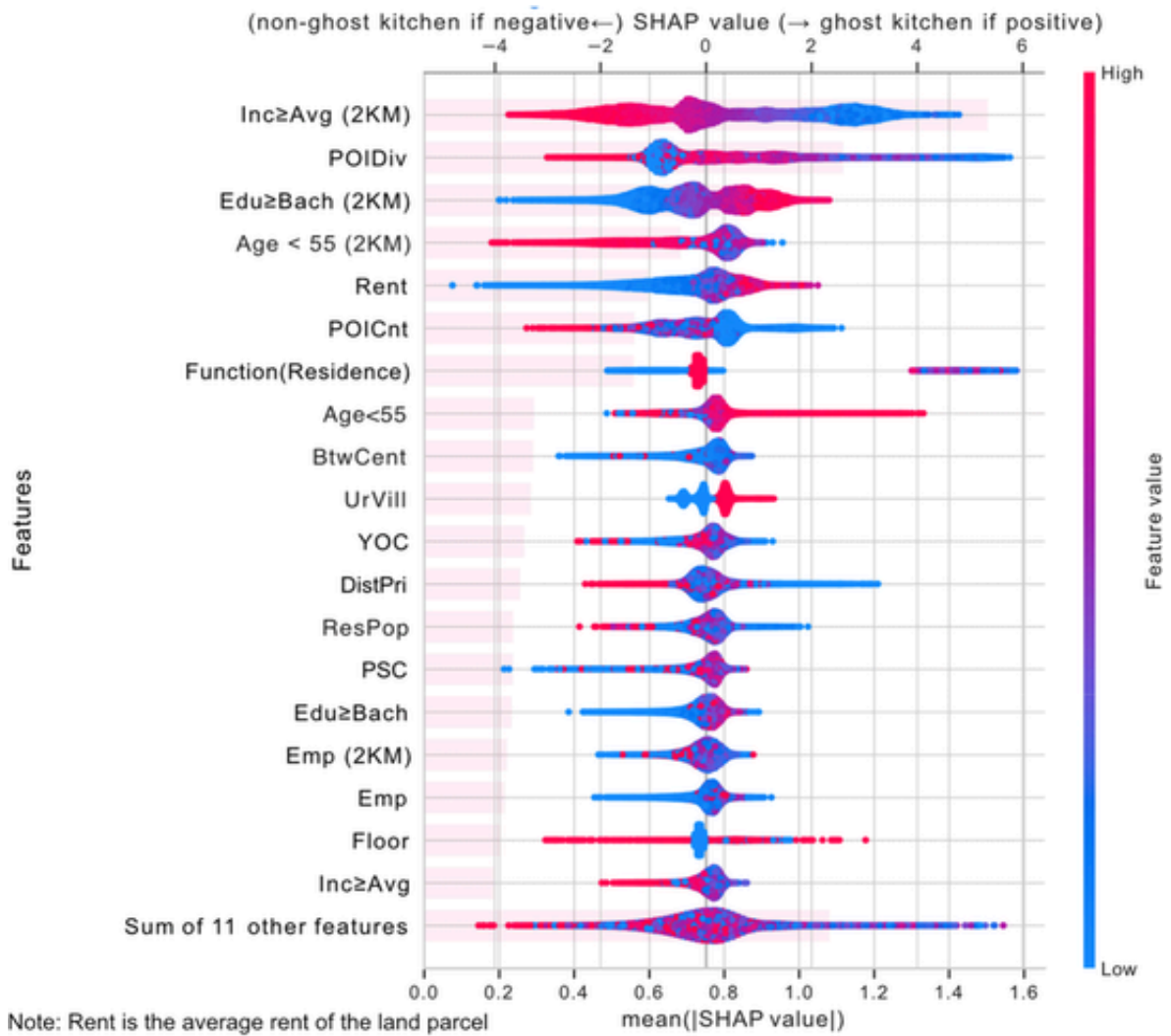


Fig. 5. Feature importance scores and SHAP values for the top 20 most influential features.

C) Built environment characteristics and delivery access

Built environment and delivery access refine this profile. Pervious surface coverage exhibits an inverted-U relationship, with the highest probabilities at moderate levels and lower probabilities in both the most intensely built blocks and the greenest, low-intensity settings (Fig. 5m). Betweenness centrality representing local network connectivity, exhibits a predominantly negative correlation with ghost kitchen presence (except in extremely low-connectivity areas) (Fig. 6h). This aligns with assumptions proposed by Vu et al. (2023) and Shapiro (2023) that prioritizing delivery efficiency represents a key location strategy differentiating ghost kitchens from conventional restaurants. Distance to primary network reveals a bimodal distribution, with higher probabilities of ghost kitchen detection either within 5 m of networks (maximizing delivery efficiency) or between 40 and 60 m (reducing visibility) (Fig. 6k). Floor level shows a pronounced V-shape, with the lowest probability at ground level and higher probabilities on basement and lower upper floors (Fig. 6o). These patterns indicate locations that are easy to serve by delivery couriers yet remain less visible to walk-by customers and routine inspections.

These findings collectively illuminate how ghost kitchens constitute a distinctive spatial phenomenon within the urban foodscape, characterized by strategic invisibility and economic pragmatism. Their loca-

tion patterns reveal a sophisticated balancing act between economic viability, operational efficiency, and deliberate inconspicuousness. By positioning themselves in economically vibrant yet functionally homogeneous areas, often within older buildings on less visible environment, ghost kitchens create a parallel food ecosystem that operates largely outside traditional consumer awareness. This shadow network capitalizes on digital connectivity while minimizing physical visibility (Graham, 2020), creating what might be termed 'digital visibility within physical obscurity'. Beyond practical implications for urban planning and food safety regulation, these patterns suggest an emerging paradigm in commercial food services that decouples consumption from place and reconfigures urban food geographies through technological mediation. As cities continue to evolve alongside digital platforms, understanding these hidden geographies becomes essential for comprehensive urban governance, especially for food safety regulations implementation, and the equitable and healthy urban foodscape development.

5. Regulatory and food safety implications of 'weak' locations

The secrecy-intended location preference of ghost kitchens renders them largely invisible to traditional public surveillance and creates significant challenges for official inspection (Y.-A. Kim and Hipp, 2021;

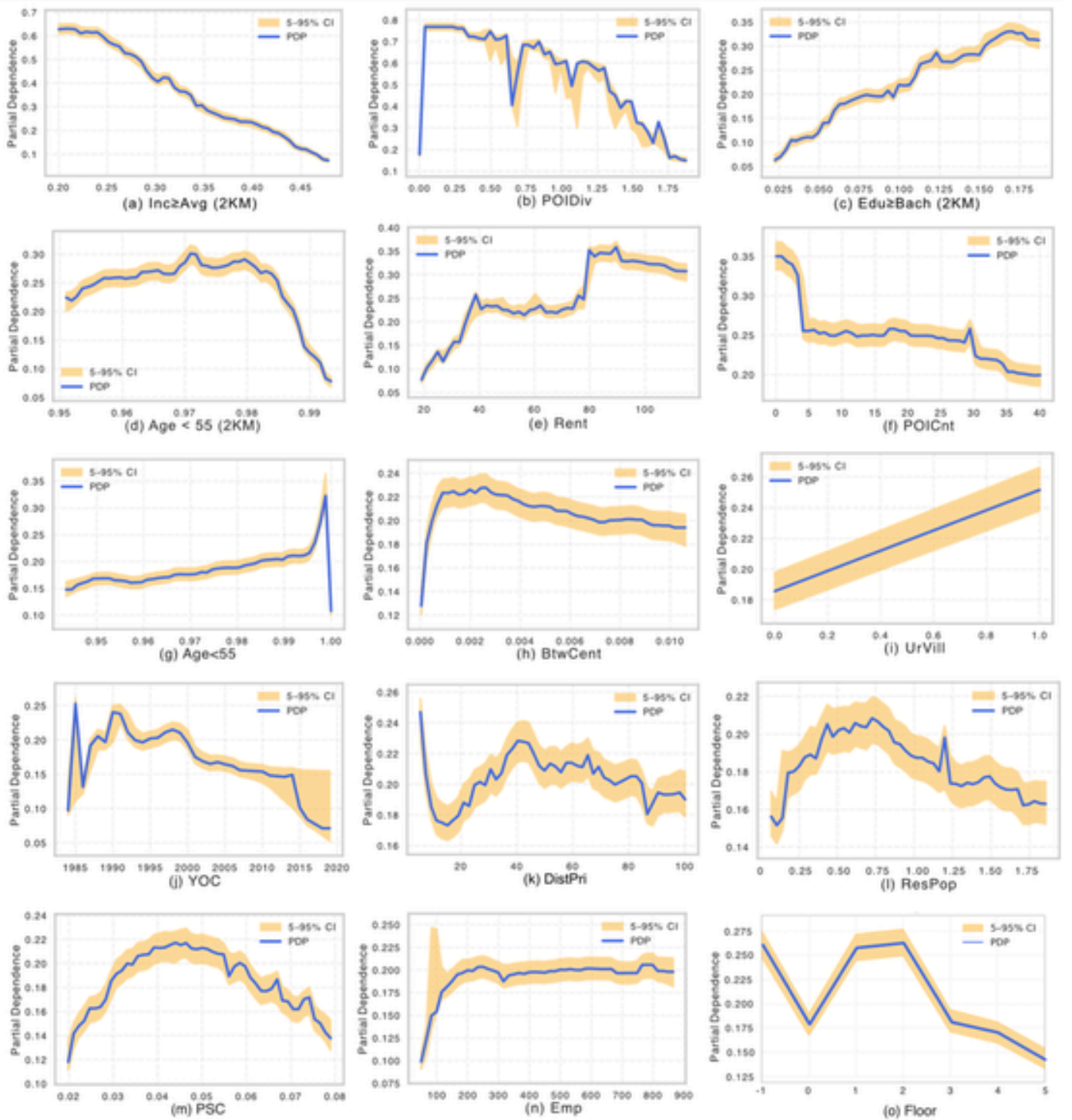


Fig. 6. Partial dependence plots of numeric predictors with 90 % percentile-bootstrap confidence intervals (5th–95th percentiles, 5-95% CI).

Laheri et al., 2025). End-users can rarely distinguish potential health risks by examining the often delicate packaging alone. Ghost kitchens strategically position themselves in 'advantageous' locations that circumvent rigid regulations, posing challenges for local authorities to identify their precise locations and requiring increased resources for site inspections due to their relatively sparse or flexible location strategies (Laheri et al., 2025). In the realm of food safety, thought food safety and related public health issue has been strengthen through monitor and surveillance policies (Wu and Chen, 2018), heightened vigi-

lance or even fraud anticipation is warranted, given historical evidence that food-related fraud tends to proliferate in less regulated environments (Dai and Wu, 2023; Laheri et al., 2025). While not assuming all such establishments engage in fraudulent practices, their location strategies inadvertently align with enablers identified in the crime triangle from criminology (Cane and Primrose, 2021): victim (consumers prioritizing convenience over verification), guardian (local authorities and platforms facing inspection difficulties), and offender (establishments potentially engaging in adulteration for higher profit margins

amidst fierce competition and reduced oversight). As any element of this triangle intensifies, the fraud opportunity correspondingly increases.

Confronting these escalating risks to food standards and public health, traditional regulatory approaches appear increasingly inadequate and require adaptation for effective digital governance. More comprehensive oversight mechanisms might include regulated platforms mandating enhanced information disclosure (e.g., clear labeling of ghost kitchen operations or live streaming of food preparation processes), incentivized reporting systems that empower delivery workers and staff to function as surveillance agents and standardized architectural protocols that require ghost kitchens to operate with greater structural transparency. Such multi-faceted approaches acknowledge the unique challenges posed by these digitally-mediated, physically obscured food establishments while balancing consumer convenience with public health imperatives.

It is important to acknowledge limitations in our analysis. Despite its comprehensiveness, our study may not capture all ghost kitchens operating in Shenzhen. While the two major platforms analyzed cover more than 95% of the total market share in online food delivery, smaller local platforms (such as Shenxin) and recently newly established JD and Taobao may also list similar restaurant types. Additionally, our research lacks precise indicators for measuring financial efficiency (such as establishment-specific rent costs and operational space requirements). These indicators could help determine whether vicinity-intended location strategies are primarily motivated by financial considerations or simply represent a strategic 'retreat into the crowd'. Future research could employ qualitative methods such as interviews, combined with detailed establishment-level data, to examine causality for more accurate inference and prediction, ultimately supporting enhanced digital governance of the urban foodscape.

By linking empirical findings to both location theory and urban governance debates, this section highlights how digital-era business models demand new forms of regulatory imagination. Ghost kitchens exemplify how platform-mediated services disrupt traditional frameworks of inspection and surveillance, creating an urgent need for adaptive governance that balances consumer convenience with public health protections.

6. Conclusions

This study has systematically unveiled the hidden geography of ghost kitchens within the urban foodscape of Shenzhen, revealing distinct spatial patterns that challenge conventional location theories. Our findings demonstrate that ghost kitchens constitute a fundamentally different spatial phenomenon compared to traditional and hybrid restaurants, employing strategic invisibility while maintaining delivery efficiency. The empirical evidence shows that ghost kitchens deliberately seek what we term 'weak' locations: areas with reduced POI diversity, older buildings, residential settings, and less visible floor positions. These choices reduce storefront costs and sustain rapid distribution to nearby demand, especially in catchments with lower income but higher educational attainment.

The spatial analysis revealed significant clustering of ghost kitchens in economically developed areas with relatively homogeneous commercial functions, contradicting traditional assumptions that restaurants universally benefit from heterogeneous urban environments. The micro-level model identified a complex set of location determinants, led by socio-demographic structure, followed by urban function and rents, and then network position with clear non-linearities. Together, these patterns depict a delivery-oriented strategy of digital visibility with physical obscurity.

The results call for a recalibration of urban foodscape theory to accommodate digital-era food services that operate under fundamentally different spatial logics. Frameworks that emphasize storefront expo-

sure, pedestrian flows, and agglomeration effects only partially explain ghost-kitchen siting, which instead balances targeted demand, operational efficiency, and strategic inconspicuousness. This is not merely a digital add-on to traditional restaurants but a structural shift that loosens the bond between consumption and place and reconfigures urban commercial space.

Policy and governance need to adapt accordingly. Conventional inspection regimes designed for street-facing premises are ill-matched to businesses that minimize physical visibility. Risk-based inspection, data-sharing and traceability mechanisms with platforms, and design standards that preserve transparency without stifling innovation can better safeguard food safety in an increasingly digital foodscape.

Finally, this work bridges location theory and platform urbanism by offering a multi-level, explainable modeling framework for hidden amenities. Future research should track temporal evolution, compare cities and cuisine types, integrate platform operations data, and employ stronger causal designs to test mechanisms, with the goal of informing adaptive, proportionate regulation that balances innovation with public health.

CRediT authorship contribution statement

Weipeng Deng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yihong Tang:** Writing – review & editing, Validation, Software, Data curation. **Yaxin Liu:** Writing – review & editing, Visualization, Data curation. **Tianren Yang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors gratefully acknowledge funding support from the National Natural Science Foundation of China (52408095), the Hong Kong Research Grants Council (17601425; 27601324), Guangdong Basic and Applied Basic Research Foundation (2025A1515010902), and the Young Elite Scientists Sponsorship Program by China Association for Science and Technology (2022ONRC001).

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jretconser.2026.104766>.

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